

Literature Status: Isotropy in the Finite-Perimeter Decomposition Theorem

arXiv:1907.10869 answered by arXiv:2103.14459

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Status. `literature_already_answered`. The later paper removes exactly the hypothesis queried by the source; this packet records the identification and does not claim a new proof.

The source question

Bonicatto–Pasqualetto–Rajala work on a PI space $(X, d, \mathfrak{m})$, meaning a complete metric measure space with doubling measure and a weak $(1, 1)$ -Poincaré inequality. A finite-perimeter set E is indecomposable if it has no disjoint positive-measure partition $E = F \cup G$ satisfying

$$P(E, X) = P(F, X) + P(G, X).$$

Their Theorem 2.14 proves, under an additional *isotropy* assumption, a unique finite or countable perimeter-additive partition of every E into maximal indecomposable pieces. Immediately before stating their results, the introduction asks whether isotropy is in fact needed for the decomposition to hold.

Exact later answer

Lahti’s arXiv:2103.14459 is titled *A note on indecomposable sets of finite perimeter*. Its abstract says that the isotropy condition can be removed. Theorem 1.1 states:

Let $(X, d, \mathfrak{m})$ be a PI space and let $E \subset X$ have finite perimeter. Then there is a unique finite or countable partition $\{E_i\}_{i \in I}$ of E into indecomposable subsets such that $\mathfrak{m}(E_i) > 0$ and

$$P(E, X) = \sum_{i \in I} P(E_i, X).$$

Uniqueness is modulo $\mathfrak{m}$ -null sets. Moreover, the E_i are maximal: every finite-perimeter indecomposable $F \subset E$ is contained modulo null sets in a unique E_i .

This reproduces every clause of the source’s Theorem 2.14 while dropping isotropy. Hence the source question has a complete affirmative answer:

isotropy is not needed for the decomposition theorem.

Why the later proof applies

The obstacle in the source proof was a hereditary additivity lemma obtained from the Hausdorff-density representation of perimeter and isotropicity. Lahti replaces it by two results proved directly from BV relaxation and weak upper gradients:

1. If $F \cap G$ is null and perimeter is additive for $F \cup G$ on a set D , then it is additive for every finite-perimeter pair $F' \subset F$, $G' \subset G$ on D .
2. If perimeter is additive for disjoint F, G on an open set W , it remains additive on every smaller open set $W' \subset W$.

The first recovers the partition-refinement lemma and stability of increasing unions of indecomposable sets. The second recovers local-to-global indecomposability. Lahti then follows the source's decomposition proof almost verbatim. No isotropicity is reintroduced elsewhere.

Scope boundary

The answer concerns the decomposition theorem, exactly as asked. The later paper notes that some other assertions in the 2020 paper, particularly about simple sets, can fail without isotropicity or an even stronger two-sided property. Those separate assertions are not promoted here.

References

- [1] P. Bonicatto, E. Pasqualetto, and T. Rajala, *Indecomposable sets of finite perimeter in doubling metric measure spaces*, arXiv:1907.10869; Calc. Var. Partial Differential Equations 59 (2020), 63.
- [2] P. Lahti, *A note on indecomposable sets of finite perimeter*, arXiv:2103.14459; Adv. Calc. Var. 16 (2023), 559–570, Theorem 1.1 and Propositions 3.1–3.2.